EXPERIMENT 1

CODE:

```
 ≡  File  Edit  Search  Run  Compile  Debug  Project  Options    Wi
[■]                          DSA1.C
#include<stdio.h>
#define MAX 10
int stack[MAX];

int top = -1;
int i;
void push(int alert){
if(top==MAX-1){
printf("Stack Overflow!\n");
}else{
stack[++top]=alert;
}
}

void pop(){
if(top==-1){
printf("No active alerts.\n");
}else{
top--;
}
}
 *      11:36
F1 Help  Alt-F8 Next Msg  Alt-F7 Prev Msg  Alt-F9 Compile  F9 Make
```

```
[■]

void display(){
if (top==-1){
printf("No active alerts.\n");
}else{
printf("Active alerts.\n");

for(i=top;i>=0;i--){
if (stack[i]==1)
printf("Yellow Alert\n");
else if (stack[i]==2)
printf("Orange Alert\n");
else if(stack[i]==3)
printf("Red Alert\n");
}
}
}
```

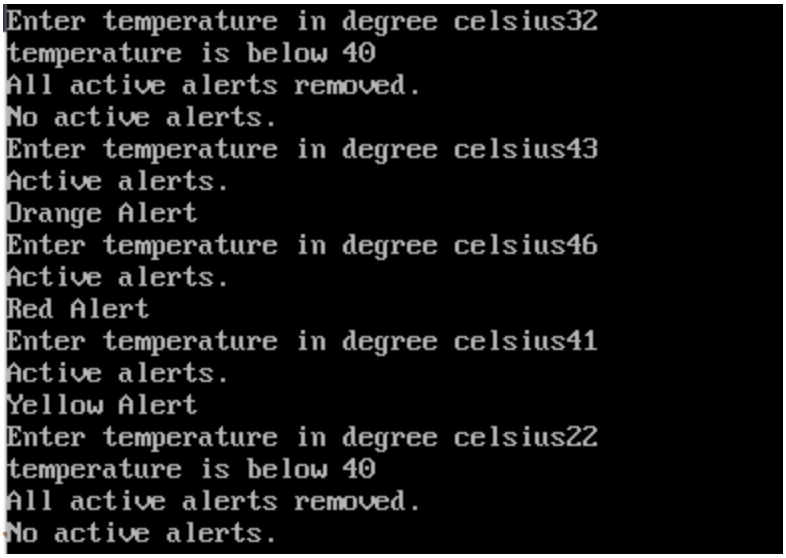
```c
int main(){
int temp;

printf("Enter temperature in degree celsius");
scanf("%d",&temp);

if (temp>=45){
push(3);
}
else if (temp>=43)
{
push(2);
}
else if(temp>=40)
{
push(1);
}
else{
printf("temperature is below 40\n");
top=-1;
printf("All active alerts removed.\n");
```

```c
if (temp>=45){
push(3);
}
else if (temp>=43)
{
push(2);
}
else if(temp>=40)
{
push(1);
}
else{
printf("temperature is below 40\n");
top=-1;
printf("All active alerts removed.\n");
}
display();
getch();
return 0;
}
```

OUTPUT:


Enter temperature in degree celsius32
temperature is below 40
All active alerts removed.
No active alerts.
Enter temperature in degree celsius43
Active alerts.
Orange Alert
Enter temperature in degree celsius46
Active alerts.
Red Alert
Enter temperature in degree celsius41
Active alerts.
Yellow Alert
Enter temperature in degree celsius22
temperature is below 40
All active alerts removed.
No active alerts.